

A tour of the model zoo

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1 Overview

`EcologicalBayesianNetworks.jl` is a curated zoo of publicly downloadable ecological Bayesian belief networks and influence diagrams. Every model has a manifest (`models/<name>/metadata.toml`) recording where it came from, how to cite it, its licence and a checksum. A permissive licence (CC BY, CC BY-SA, CC0, MIT) is what *permits* shipping the original file with the package, but it does not by itself decide: each manifest's `redistribution` field does, and several permissively licensed records are still `fetch-only` or `reconstruction` because of how the source publishes them. Records that are not shipped verbatim are fetched into a local cache on demand. This vignette walks through the catalogue, loads a benchmark network, solves a decision model, and computes a marginal on a small chance-only network.

The models are real ones: belief networks published for habitat, conservation and natural-resource problems (McCann et al. 2006; Newton et al. 2007), environmental risk assessments (Moe et al. 2021; Kaikkonen et al. 2021) and ecosystem-service models (Landuyt et al. 2013). They come from four kinds of source – the Bayesian Network Model Archive (BNMA 2026), the bnlearn network repository (Scutari 2010), the bnRep collection (Leonelli 2025), and the supplementary material of individual papers – and each brings its own licence, which is why the zoo enforces a redistribution policy rather than mirroring everything.

2 Setup

```
using EcologicalBayesianNetworks
n_models()
```

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3 The catalogue

`model_catalog()` returns a filtered view of the registry that prints as a table. The `kind` column says whether a model is a plain Bayesian network (BN), has decision or utility nodes (ID, an influence diagram), or is a package: a catalogue record for a Dryad, Zenodo, figshare or source-repository archive from which a network can be rebuilt but which is not itself a network file.

```
model_catalog()
```

ModelCatalog with 41 models

name	category	format	licence
redistribution nodes arcs kind			
reference_grazing_id	reference	julia	MIT
builtin 12 13 ID			
reference_habitat_bn	reference	julia	MIT
builtin 7 6 BN			
beach_mice_bn	habitat	neta	CC0 1.0 (ScienceBase record);
US Government work in the public domain		fetch-only	0 0 BN
brown_bear_recreation	habitat	dne	CC-BY-NC-ND (version
unstated)			fetch-only 37 45
BN			
gulf_sturgeon_bn	habitat	neta	CC0 1.0 (ScienceBase record);
US Government work in the public domain		fetch-only	0 0 BN
habitat_suitability_tiger	habitat	xdsl	CC-BY (version unstated)
verbatim 11 12 BN			

plexus_teb_bat_site			habitat	dne	unstated		
fetch-only	13	15	BN				
plexus_teb_bat_subwatershed			habitat	dne	unstated		
fetch-only	7	6	BN				
seagrass_dbn_package			habitat	package	Dryad record terms (consult the record; Dryad lists its content as reusable)	reconstruction	0 0
package							
thailand_vertebrate_bbn_package			habitat	package	figshare collection terms (consult the individual items)	reconstruction	0 0
package							
bull_trout_food_web			population	dne	CC-BY-NC-ND (version unstated)	fetch-only	10 15
BN							
marten_age			population	dne	CC-BY (version unstated)		
fetch-only	7	6	BN				
marten_telomere			population	dne	CC-BY-NC-ND (version unstated)	fetch-only	0 0
BN							
native_fish_v1			population	dne	CC-BY (version unstated)		
verbatim	7	8	BN				
pacific_walrus			population	dne	CC-BY-NC-ND (version unstated)	fetch-only	77 96
BN							
plexus_population_viability			population	dne	unstated		
fetch-only	11	13	ID				
plexus_teb_bat_basin			population	dne	unstated		
fetch-only	8	7	BN				
polar_bear_stressor_i			population	dne	CC-BY-NC-ND (version unstated)	fetch-only	38 44
BN							
polar_bear_stressor_ii			population	dne	CC-BY-NC-ND (version unstated)	fetch-only	49 53
BN							
ballycanew_diffuse_p			water	xdsl	MIT		
verbatim	4	2	BN				
ballycanew_point_diffuse_p			water	xdsl	MIT		
verbatim	12	8	BN				
bnma_water			water	dne	CC-BY-SA (version unstated)		
verbatim	32	66	BN				
vansjo_seasonal_bn_package			water	package	unstated (repository states no licence; check the repository terms)	reconstruction	0 0
package							
shark_management_survey			risk	dne	CC-BY-NC (version unstated)		
fetch-only	26	54	BN				
tidal_saline_wetlands			risk	dne	CC-BY-ND (version unstated)		
fetch-only	18	16	BN				
arhopalus_flight_activity			biosecurity	dne	CC-BY-NC-ND (version unstated)	fetch-only	7 11
BN							
fisram_freshwater_species			biosecurity	dne	CC-BY-ND (version unstated)		
fetch-only	20	23	BN				

hylastes_flight_activity			biosecurity	dne	CC-BY-NC-ND (version		
unstated)					fetch-only	7	11
BN							
hylurgus_flight_activity			biosecurity	dne	CC-BY-NC-ND (version		
unstated)					fetch-only	10	17
BN							
microctonus_ng_risk			biosecurity	xdsl	CC-BY-NC-ND (version		
unstated)					fetch-only	45	51
BN							
wild_pig_hsi_package			biosecurity	package	CC0 1.0		
reconstruction	0	0	package				
brown_trout_bdn			decision	dne	CC-BY (version unstated)		
verbatim	9	13	ID				
koalas			decision	dne	CC-BY (version unstated)		
verbatim	10	11	ID				
song_sparrow_bdn			decision	dne	CC-BY (version unstated)		
verbatim	12	15	ID				
waterhole_fence			decision	dne	CC-BY (version unstated)		
verbatim	10	12	ID				
animals			benchmark	dne	CC-BY-NC-ND (version		
unstated)					fetch-only	7	6
BN							
barley			benchmark	net	CC BY-SA 3.0		
verbatim	48	84	BN				
bnrep_package			benchmark	package	MIT		
reconstruction	0	0	package				
hailfinder			benchmark	net	CC BY-SA 3.0		
verbatim	56	66	BN				
mildew			benchmark	net	CC BY-SA 3.0		
verbatim	35	46	BN				
water			benchmark	net	CC BY-SA 3.0		
verbatim	32	66	BN				

Categories, in display order:

```
model_categories()
```

```
8-element Vector{String}:
```

```
"reference"
"habitat"
"population"
"water"
"risk"
"biosecurity"
"decision"
"benchmark"
```

Filters combine: the benchmark networks, or only the influence diagrams whose files ship with the package.

```
available_models(category = "benchmark")
```

```
6-element Vector{String}:  
"animals"  
"barley"  
"bnrep_package"  
"hailfinder"  
"mildew"  
"water"
```

```
available_models(kind = :id, redistributable = true)
```

```
5-element Vector{String}:  
"brown_trout_bdn"  
"koalas"  
"reference_grazing_id"  
"song_sparrow_bdn"  
"waterhole_fence"
```

`model_info` looks a model up by name, title or file name, ignoring case and punctuation, and returns its `ModelSpec`.

```
model_info("Song Sparrow BDN")
```

```
ModelSpec song_sparrow_bdn  
  title          Song Sparrow riparian restoration decision network  
  (SongSparrowBDN_v250804)  
  category       decision  
  tags           bird, riparian, restoration, mercury, decision network, BNMA  
  format         dne  
  file           SongSparrowBDN_v250804.dne  
  licence        CC-BY (version unstated)  
  redistribution  verbatim  
  nodes / arcs   12 / 15  
  kind           influence diagram  
  source         https://bnma.co/bn/2893  
  download       https://bnma.co/bnrepo/?slice=download&format=dne&bnId=2893  
  doi            10.59381/ofwegrfaqyo  
  citation       Kotalik, C. J., Rowland, F. E., Marcot, B. G., Skrabis, K.,  
  Walters, D. M., Hinck, J. E., Clements, W., Richer, E. and Isanhart, J. (2026).  
  Causal networks to inform decisions for ecological restoration. Environmental  
  Management 76(7).  
  retrieved      2026-09-06  
  sha256         9be7fb3ba119d0f5c8a00509fe87d19827043b6ac26d7dce5b57c679c7181666  
  parser options allow_missing_tables = true, strict = true
```

notes Original .dne as served by the BNMA converter (BN DOI 10.59381/ofwegrfqyo); filename unchanged. Decision network with planting and soil-treatment decisions and utility nodes (8 chance, 2 decision, 2 utility nodes; 7 CONSTANT nodes excluded); load_model returns an InfluenceDiagramModel whose two table-less mechanisms stay unbound, so it cannot be optimised. Two chance nodes (RiparianPlantCost_infl_adj, Time) are Netica finding/equation nodes with evidence but no CPT, so the manifest sets allow_missing_tables = true and model_ir validates with BayesianNetworkFormats.validate(ir; allow_missing_tables = true).

4 Licences and redistribution

The manifest redistribution field is the licence policy in one word:

- verbatim: the original file is committed next to a LICENSE.txt with the attribution (the CC BY-SA 3.0 benchmarks of bnlearn ([Scutari 2010](#)), the MIT-licensed Ballycanew phosphorus tools, the CC BY records of the BNMA repository ([BNMA 2026](#)), the CC BY networks collected in bnRep ([Leonelli 2025](#)));
- fetch-only: metadata only. No-derivatives, non-commercial and unstated licences are never committed and never converted; fetch_model downloads them into a scratch cache when a direct URL exists, and import_model copies a file you downloaded by hand;
- builtin: built in Julia (the reference models of the SPEC);
- reconstruction: a data or code archive rather than a network. Nothing is committed and nothing is cached; load_model raises ReconstructionOnlyError pointing at the record.

```
model_catalog(kind = :package)
```

ModelCatalog with 5 models

name	category	format	licence
redistribution nodes arcs kind			
-----	-----	-----	
-----	-----	-----	
seagrass_dbn_package	habitat	package	Dryad record terms (consult the record; Dryad lists its content as reusable) reconstruction 0 0
thailand_vertibrate_bbn_package	habitat	package	figshare collection terms (consult the individual items) reconstruction 0 0
vansjo_seasonal_bn_package	water	package	unstated (repository states no licence; check the repository terms) reconstruction 0 0
wild_pig_hsi_package	biosecurity	package	CC0 1.0
bnrep_package	benchmark	package	MIT

```
try
  load_model("bnrep_package")
catch e
  println(sprint(showerror, e))
end
```

ReconstructionOnlyError: model "bnrep_package" is a catalogue record for a data or code archive, not a network file, so there is nothing to load. The archive is at <https://github.com/manueleleonelli/bnRep>. Licence: MIT. Rebuild the network from the archive yourself, then register it as a model of its own with `import_model`.

```
model_catalog(redistributable = false)
```

ModelCatalog with 26 models

name	category	format	licence
redistribution nodes arcs kind			
-----	-----	-----	
-----	-----	-----	
beach_mice_bn	habitat	neta	CC0 1.0 (ScienceBase record);
US Government work in the public domain		fetch-only	0 0 BN
brown_bear_recreation	habitat	dne	CC-BY-NC-ND (version
unstated)			fetch-only 37 45
BN			
gulf_sturgeon_bn	habitat	neta	CC0 1.0 (ScienceBase record);
US Government work in the public domain		fetch-only	0 0 BN
plexus_teb_bat_site	habitat	dne	unstated
fetch-only 13 15 BN			
plexus_teb_bat_subwatershed	habitat	dne	unstated
fetch-only 7 6 BN			
seagrass_dbn_package	habitat	package	Dryad record terms (consult
the record; Dryad lists its content as reusable)		reconstruction	0 0
package			
thailand_vertibrate_bbn_package	habitat	package	figshare collection terms
(consult the individual items)		reconstruction	0 0
package			
bull_trout_food_web	population	dne	CC-BY-NC-ND (version
unstated)			fetch-only 10 15
BN			
marten_age	population	dne	CC-BY (version unstated)
fetch-only 7 6 BN			
marten_telomere	population	dne	CC-BY-NC-ND (version
unstated)			fetch-only 0 0
BN			
pacific_walrus	population	dne	CC-BY-NC-ND (version
unstated)			fetch-only 77 96
BN			
plexus_population_viability	population	dne	unstated
fetch-only 11 13 ID			
plexus_teb_bat_basin	population	dne	unstated
fetch-only 8 7 BN			

polar_bear_stressor_i unstated)	population	dne	CC-BY-NC-ND (version fetch-only	38	44
BN					
polar_bear_stressor_ii unstated)	population	dne	CC-BY-NC-ND (version fetch-only	49	53
BN					
vansjo_seasonal_bn_package no licence; check the repository terms)	water	package	unstated (repository states reconstruction	0	0
package					
shark_management_survey fetch-only	26	54	BN	risk	dne
tidal_saline_wetlands fetch-only	18	16	BN	risk	dne
arhopalus_flight_activity unstated)	biosecurity	dne	CC-BY-NC-ND (version fetch-only	7	11
BN					
fisram_freshwater_species fetch-only	20	23	BN	biosecurity	dne
hylastes_flight_activity unstated)	biosecurity	dne	CC-BY-NC-ND (version fetch-only	7	11
BN					
hylurgus_flight_activity unstated)	biosecurity	dne	CC-BY-NC-ND (version fetch-only	10	17
BN					
microctonus_ng_risk unstated)	biosecurity	xdsl	CC-BY-NC-ND (version fetch-only	45	51
BN					
wild_pig_hsi_package reconstruction	0	0	package	biosecurity	package
animals unstated)			package	benchmark	dne
BN					
bnrep_package reconstruction	0	0	package	benchmark	package
					MIT

Every verbatim model carries its licence text:

```
print(licence_text("water"))
```

This network file is redistributed verbatim from the bnlearn Bayesian Network Repository (<https://www.bnlearn.com/bnrepository/>), maintained by Marco Scutari, under the Creative Commons Attribution-Share Alike 3.0 Unported licence (CC BY-SA 3.0, <https://creativecommons.org/licenses/by-sa/3.0/>).

Attribution: Marco Scutari, bnlearn Bayesian Network Repository.
Scutari, M. (2010). Learning Bayesian Networks with the bnlearn R Package. Journal of Statistical Software, 35(3), 1-22. <https://doi.org/10.18637/jss.v035.i03>

The file is unmodified. Derivative works must be distributed under the same licence.

5 Loading a benchmark network

`water` (Jensen et al. 1989, from the bnlearn repository ([Scutari 2010](#))) is stored gzipped exactly as served; the loader decompresses it in memory, applies the manifest's parser options and hands the `NetworkIR` to `BayesianNetworks.BayesModel`.

```
water = load_model("water")
```

```
BayesModel(32 variables, 32 mechanisms, 32 kernels)
```

```
using BayesianNetworks: validate
validate(water; semantics = true)
```

`model_summary` parses a model and counts what is in it without building a `BayesModel`, so it also works for influence diagrams:

```
model_summary("water")
```

```
ModelSummary water (WATER: waste-water treatment plant control)
  format / licence      net / CC BY-SA 3.0 (verbatim)
  nodes                 32 (32 chance, 0 decision, 0 utility)
  arcs                  66
  largest state space    4
  largest in-degree      5
  load_model             BayesModel
```

```
model_summary("waterhole_fence")
```

```
ModelSummary waterhole_fence (Waterhole Fence: fencing and plant survival decision
network)
  format / licence      dne / CC-BY (version unstated) (verbatim)
  nodes                 10 (6 chance, 1 decision, 3 utility)
  arcs                  12
  largest state space    3
  largest in-degree      3
  load_model             InfluenceDiagramModel
```

Decision models load as `InfluenceDiagrams.InfluenceDiagramModels`, with their chance kernels and tabular utilities bound and the decision nodes' parents as information sets:

```
koalas = load_model("koalas")
```

```
InfluenceDiagramModel(6 variables, 2 decisions, 4 utilities, 4 kernels, 4 bound)
```

```
using InfluenceDiagrams: optimize, policy_table
sol = optimize(koalas)
sol.expected_utility, policy_table(sol.strategy[:Cull]),
policy_table(sol.strategy[:Relocate])
```

```
(20.199999999999996, fill(:No), [:Yes, :Yes, :Yes])
```

The NetworkIR behind any model is one call away:

```
[(v.id, v.kind, length(v.parents)) for v in model_ir("koalas").variables]
```

```
10-element Vector{Tuple{Symbol, BayesianNetworkFormats.NodeKind, Int64}}:
 (:Cull, BayesianNetworkFormats.DecisionNode, 0)
 (:Relocate, BayesianNetworkFormats.DecisionNode, 1)
 (:KoalaPopulation, BayesianNetworkFormats.ChanceNode, 2)
 (:U, BayesianNetworkFormats.UtilityNode, 1)
 (:NveSocialImplications, BayesianNetworkFormats.ChanceNode, 2)
 (:V, BayesianNetworkFormats.UtilityNode, 1)
 (:TreeSpecies, BayesianNetworkFormats.ChanceNode, 1)
 (:Tourists, BayesianNetworkFormats.ChanceNode, 1)
 (:W, BayesianNetworkFormats.UtilityNode, 1)
 (:X, BayesianNetworkFormats.UtilityNode, 1)
```

6 A marginal on a small network

native_fish_v1 (Bayesian Intelligence, CC BY) links rainfall, drought, pesticide use and river flow to native fish abundance. With seven nodes the reference joint distribution is tiny, so BayesianNetworks.marginal answers exactly.

```
using BayesianNetworks: marginal, observe, probability
fish = load_model("native_fish_v1")
marginal(fish, :FishAbundance)
```

```
FiniteKernel{Float64}(I → FishAbundance{High,Medium,Low})
()
  High    0.2569
  Medium  0.2239
  Low     0.5193
```

Conditioning on a drought year:

```
marginal(observe(fish, :Drought ⇒ :Yes), :FishAbundance)
```

```
FiniteKernel{Float64}(I → FishAbundance{High,Medium,Low})
()
  High    0.1288
  Medium  0.1756
  Low     0.6957
```

```
[(s, round(probability(marginal(observe(fish, :Drought ⇒ :Yes), :FishAbundance),
s); digits = 3))
 for s in (:High, :Medium, :Low)]
```

```
3-element Vector{Tuple{Symbol, Float64}}:
 (:High, 0.129)
 (:Medium, 0.176)
 (:Low, 0.696)
```

7 The fetch-only workflow

Fetch-only models have a `download_url` (when the host serves the file directly) and `fetch_instructions` for the manual route. Nothing below touches the network: the calls that would be shown, not run.

```
spec = model_info("tidal_saline_wetlands")
(spec.licence, spec.redistribution, spec.download_url)

("CC-BY-ND (version unstated)", "fetch-only",
 "https://bnma.co/bnrepo/?slice=download&format=dne&bnId=209")

println(spec.fetch_instructions)
```

Open the BNMA export page (`source_url` with `body=convert`), choose 'Netica .dne format', press Save, and then `import_model(name, path)`; or call `fetch_model(name)`, which uses the same GET endpoint the Save button uses. The licence forbids redistribution of the file, so it is never committed.

Asking for the file before it is fetched raises a `ModelNotFetchedError` whose message repeats the instructions. Rendering this vignette must not disturb whatever you have already downloaded, so the demonstration runs inside `with_cache_dir`, which points `cache_dir()` at a temporary directory for the duration of the block. The error message names that directory by its absolute path, which differs from machine to machine, so the vignette prints it as `<temporary cache>`:

```
mktempdir() do tmp
  with_cache_dir(tmp) do
    try
      model_path("tidal_saline_wetlands")
    catch e
      println(replace(sprint(showerror, e), cache_dir() => "<temporary
        cache>"))
    end
  end
end
```

`ModelNotFetchedError`: the file of model "tidal_saline_wetlands" is not in the cache (<temporary cache>). Run `fetch_model("tidal_saline_wetlands")` to download it from <https://bnma.co/bnrepo/?slice=download&format=dne&bnId=209>, or obtain it by hand and call `import_model("tidal_saline_wetlands", path)`. Instructions: Open the BNMA export page (`source_url` with `body=convert`), choose 'Netica .dne format', press Save, and then `import_model(name, path)`; or call `fetch_model(name)`, which uses the same GET endpoint the Save button uses. The licence forbids redistribution of the file, so it is never committed.

`with_cache_dir` is the safe way to run anything that writes to or clears the cache (`fetch_model`, `import_model`, `clear_cache`); setting `ECOLOGICAL_BN_CACHE` does the same for a whole session. Nothing above deleted a file, and nothing below does either: the two ways to make the model available are shown, not run.

```
fetch_model("tidal_saline_wetlands")           # direct download, checksum verified
import_model("tidal_saline_wetlands", "Tidal Saline Wetlands.dne") # a file saved by
hand                                           # remove it again
clear_cache(name = "tidal_saline_wetlands")
```

After a fetch, `is_available("tidal_saline_wetlands")` is true and `model_ir` reads the file from the cache (`cache_dir()`). `verify_checksums(include_cache = true)` re-checks every committed and cached file against the manifests, and `verify_checksums(collect_mismatches = true)` reports every bad file rather than stopping at the first.

```
is_available("tidal_saline_wetlands"), length(fetchable_models())
```

```
(true, 20)
```

8 Reference models

Two models are built in Julia rather than read from files: the SPEC section 45 habitat network (`reference_habitat_bn`, structure only, and `reference_habitat_model`, with its conditional probability tables) and the section 46 grazing-management influence diagram (`reference_grazing_id`, an `InfluenceDiagramModel`). The following vignettes work through both.

```
habitat = load_model("reference_habitat_bn")
marginal(habitat, :Occupancy)
```

```
FiniteKernel{Float64}{I → Occupancy{absent,present}}
  ()
  absent  0.5238
  present 0.4762
```

```
reference_grazing_id()
```

```
InfluenceDiagramModel(10 variables, 1 decision, 2 utilities, 9 kernels, 2 bound)
```

9 Summary

The zoo is a registry of manifests: every model records its source, citation, DOI, licence, checksum and retrieval date, and the `redistribution` field decides whether the file is committed verbatim, fetched into a scratch cache on demand, built in Julia, or carried as a catalogue record only. That policy is what makes it safe to ship published networks alongside the code, and `licence_text` reproduces the terms next to every verbatim file. The next vignette, *The reference habitat network*, works one model end to end.

10 References

BNMA. 2026. *The Bayesian Network Model Archive*. <https://bnma.co/bnrepo/>.

Kaikkonen, Laura, Tuuli Parviainen, Mika Rahikainen, Laura Uusitalo, and Annukka Lehikoinen. 2021. “Bayesian Networks in Environmental Risk Assessment: A Review.” *Integrated Environmental Assessment and Management* 17 (1): 62–78. <https://doi.org/10.1002/ieam.4332>.

Landuyt, Dries, Steven Broekx, Rob D’hondt, Guy Engelen, Joris Aertsens, and Peter L. M. Goethals. 2013. “A Review of Bayesian Belief Networks in Ecosystem Service Modelling.” *Environmental Modelling & Software* 46: 1–11. <https://doi.org/10.1016/j.envsoft.2013.03.017>.

Leonelli, Manuele. 2025. *bnRep: A Repository of Bayesian Networks from the Academic Literature*. <https://github.com/manueleleonelli/bnRep>.

McCann, Robert K., Bruce G. Marcot, and Rick Ellis. 2006. “Bayesian Belief Networks: Applications in Ecology and Natural Resource Management.” *Canadian Journal of Forest Research* 36 (12): 3053–62. <https://doi.org/10.1139/x06-238>.

Moe, S. Jannicke, John F. Carriger, and Miriam Glendell. 2021. “Increased Use of Bayesian Network Models Has Improved Environmental Risk Assessments.” *Integrated Environmental Assessment and Management* 17 (1): 53–61. <https://doi.org/10.1002/ieam.4369>.

Newton, Adrian C., Gavin B. Stewart, Anita Diaz, Duncan Golicher, and Andrew S. Pullin. 2007. “Bayesian Belief Networks as a Tool for Evidence-Based Conservation Management.” *Journal for Nature Conservation* 15 (2): 144–60. <https://doi.org/10.1016/j.jnc.2007.03.001>.

Scutari, Marco. 2010. “Learning Bayesian Networks with the bnlearn R Package.” *Journal of Statistical Software* 35 (3): 1–22. <https://doi.org/10.18637/jss.v035.i03>.